

Not Steam - Final Project Report

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Problem Statement

Gamers such as ourselves often have many games across various libraries because of the multifaceted nature of the gaming industry. This leads to issues when trying to decide what to play, having to flip between various different libraries and systems in order to see what appeals to us at that moment. Our project is going to be a video game library and management system. It will be a third-party system that integrates games from Steam, Epic Games, etc. The system will keep track of game names and genres, as well as the most recently played games. Entities include players, games, game manufacturers, game origins (the store the game was imported from), genres, and time played. The database will be a desktop application. The user will manually input the game, genre, platform, etc.

Conceptual Database Design

Functions

Profile Creation:

Description:

Users can create a profile, where they can add their email and a custom username and password. Each user is assigned a unique user_id

Entities Involved - USER

Add Games:

Description:

Users can add games to their account. They will be asked to add the game, as well as assign a genre and tags to the game.

Entities Involved - USER, GAME, GENRE, TAG

Edit/Delete Games:

Description:

Users can edit the genre and tags a game has. They can also completely remove a game, which also removes that game's attributes.

Entities Involved - USER, GAME, GENRE, TAG

Search Library:

Description:

Users can search their game library based on name, genre, or tag.

Entities Involved - GAME, GENRE, TAG

View Game Information:

Description:

Users can view the information of each game in their library.

Entities Involved - GAME, GENRE, TAG

Add Games to Wishlist:

Description:

Users can add games to a custom wishlist. They do not own these games, but they do want them in the future.

Entities Involved - WISHLIST, GAME

Remove Games from Wishlist:

Description:

Users can remove games from their wishlist when they have bought the game or no longer want the game.

Entities Involved - WISHLIST, GAME

Assign Tags to Games:

Description:

Users can create custom tags and assign them to games.

Entities Involved - GAME, TAG

Assign Genres to Games:

Description:

Users can create custom genres and assign them to games.

Entities Involved - GAME, GENRE

Assign Media Type to Games:

Description:

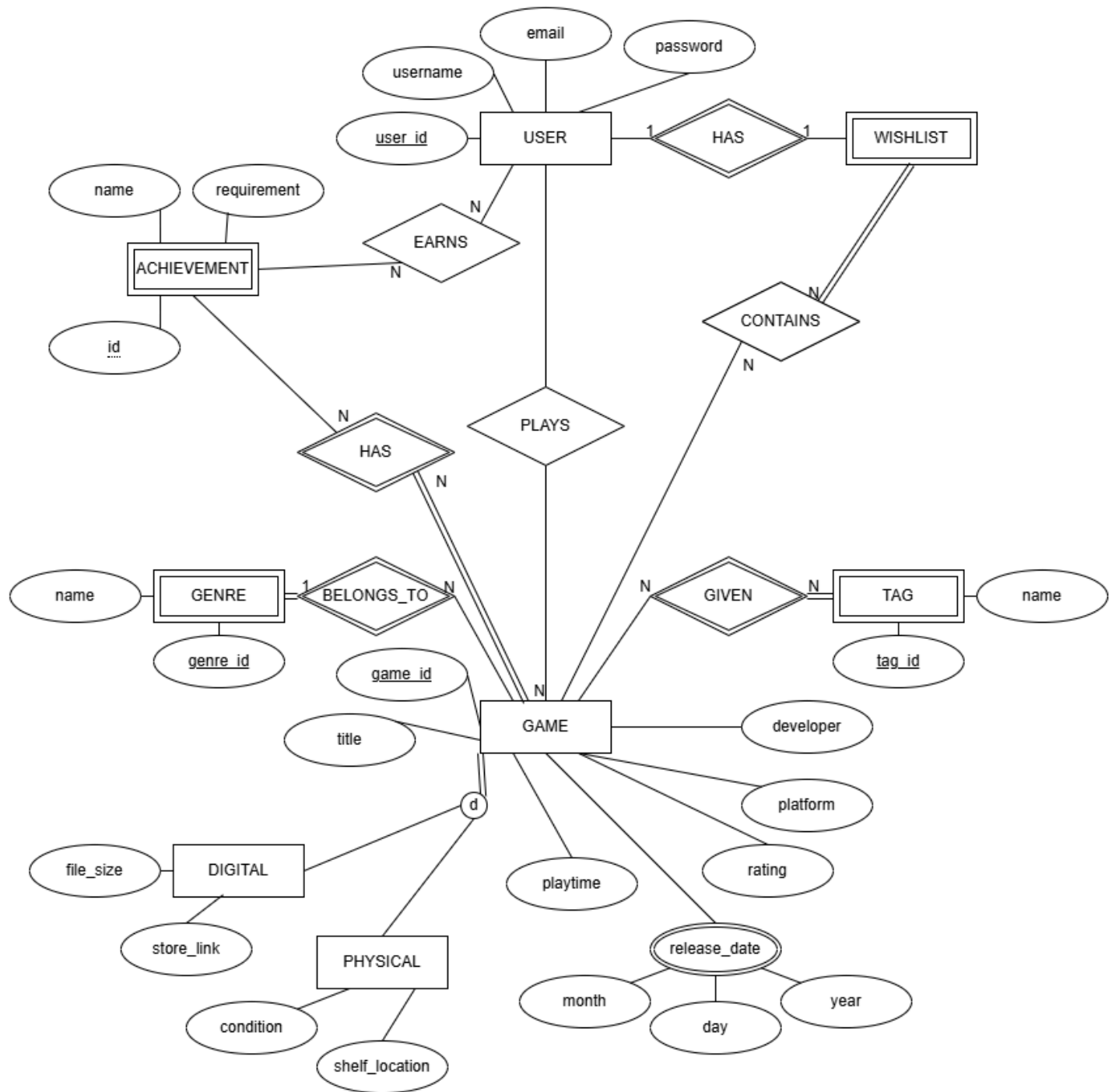
Users can assign media as digital or physical.

Entities Involved - GAME, DIGITAL, PHYSICAL

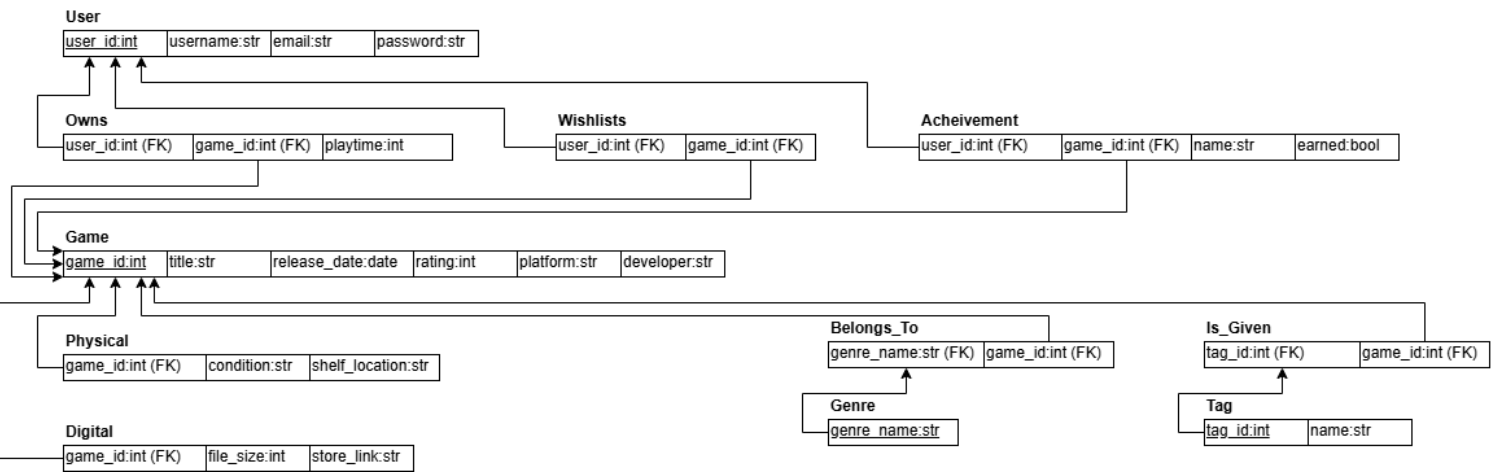
Achievements:

Users can earn achievements for games.

Entities Involved - USER, ACHIEVEMENT, GAME



Logical Database Design



Application Program Design

Create_Profile()

- name = username prompt
- if (username already exists)
 - ask for username again
- email = email prompt
- password = password prompt

Log_In()

- name = username prompt
- password = password prompt
- if (username and password don't match)
 - ask for both again

Add_Game()

- // read in file
- if(any game param besides rating is null)
 - Prompt user to correct game data
- if (title already exists)
 - Prompt user to give a different title
- Add game to owned games list

Edit_Game()

- // Uses passed parameters to edit game data
- game_to_edit = game
- if (passed_param.type in game)
 - edit game
- replace old game data with new

Delete_Game()

- game = game prompt
- remove game from library

Search_Library()

- // Depending on the search parameter, parse database for specified game
- search = search prompt

View_Game_Info()

- // Bring up all game data on the specified game
- game = game prompt

Wishlist_Game()

- // Add game to wishlist table
- // read in file
- if(any game param besides rating is null)

```
        Prompt user to correct game data
    if (title already exists)
        Prompt user to give a different title
    Add game to wishlisted games list
```

```
Remove_Wishlist_Game()
    // Remove game from wishlist table
    game = game name prompt
    remove game from wishlist
```

```
Assign_Tag()
    // Give game specified tag
    Prompt user to give game and tag
    If tag not in Tag
        Add tag to Tag
    Add tag to Game
```

```
Assign_Genre()
    // Give game specified genre
    Prompt user to give game and genre
    If the genre is not in genre
        Add genre to Genre
    Add genre to game
```

```
Assign_Game_Type()
    // Give game specified type
    Prompt user to gie game and type
    If type is Digital
        Get size and store link.
        Add to Digital for the game.
    Else
        Get condition and location
        Add to physical for game
```

```
Get_Achievement()
    // When a user obtains an achievement, add it to their achievements list
    Prompt user for game and achievement
    If achievement not in achievement for specific user and game
        Add achievement and set true
    Else
        Update achievement to true
```

Not Steam - Installation Instructions

Intended for Windows 11, Ubuntu, or macOS

Dependencies:

Python 3.10 or above, Tkinter, SQLite

For Windows:

Clone Git repo: <https://github.com/john-Garside/2300-Project>

Navigate to the repository and run the following in Powershell:

`.\setup_windows.bat`

For Ubuntu (Installation is similar for other Linux distros):

Clone Git repo: <https://github.com/john-Garside/2300-Project>

Navigate to the repository and run the following in your terminal:

`./setup_ubuntu.sh`

For Mac:

Clone Git repo: <https://github.com/john-Garside/2300-Project>

Navigate to the repository and run the following in your terminal:

`./setup_mac.sh`

Not Steam - User Manual

Command Line Interface:

To launch the CLI, navigate to the folder where you cloned the project repository. For example, my cloned folder is in ~/repos

```
maddoxreed@MaddoxPC:~/repos/2300-Project$ ls
__pycache__  database.py  gui.py  main.py  notsteam.db  setup.sql
```

For Windows, use Powershell to navigate to the correct folder. For Ubuntu and Mac, use the Terminal.

Launch main.py with the following command:

```
python3 main.py
```

The CLI should appear on screen as shown:

```
--- Not Steam ---
1. Login
2. Register
3. Quit
Choice: |
```

Select an option by typing the number next to it. For example, to log in, I will type 1:

```
--- Not Steam ---
1. Login
2. Register
3. Quit
Choice: 1|
```

Press Enter after making a choice

When logging in or registering, this will prompt you for a username. Type your username and press enter:


```
--- Not Steam ---  
1. Login  
2. Register  
3. Quit  
Choice: 1  
Username: Maddox|
```

Then, type your password and press enter again.

A menu of options will then appear, with several actions that you can perform:

```
--- Welcome, Maddox ---  
1. View my library  
2. Add a game  
3. Edit a game  
4. Delete a game  
5. Search by title  
6. Search by genre  
7. Search by tag  
8. View wishlist  
9. Add to wishlist  
10. Remove from wishlist  
11. View achievements  
12. Add achievement  
13. Assign tag to game  
14. Assign genre to game  
15. Set game as digital  
16. Set game as physical  
17. View total playtime  
18. Logout  
  
Choice: |
```

View my library

Enter 1 to view your library:

```
Choice: 1  
  
--- Your Library ---  
- Elden Ring (FromSoftware) - 444 hrs  
- Stardew Valley (ConcernedApe) - 2 hrs  
- Cuphead (Studio MDHR) - 12 hrs
```

Add a game

Enter 2 to add a game. Here is an example input for adding a game:

```
Choice: 2
Title: Helldivers 2
Release date (YYYY-MM-DD): 2024-02-08
Rating (0-10): 9
Platform: Steam
Developer: Arrowhead
'Helldivers 2' added to catalog.
Add to your library too? (y/n): y
Added to your library.
```

After adding a game, it should appear in your library:

```
Choice: 1

--- Your Library ---
- Elden Ring (FromSoftware) - 444 hrs
- Stardew Valley (ConcernedApe) - 2 hrs
- Cuphead (Studio MDHR) - 12 hrs
- Helldivers 2 (Arrowhead) - 0 hrs
```

Edit a game

Enter 3 to edit a game. Here is an example input for editing a game:

```
Choice: 3
Title of game to edit: Helldivers 2
Column to edit (title, release_date, rating, platform, developer): title
New value: Helldivers II
Game updated.
```

If name or developer is changed, it should be reflected in your library:

```
Choice: 1

--- Your Library ---
- Elden Ring (FromSoftware) - 444 hrs
- Stardew Valley (ConcernedApe) - 2 hrs
- Cuphead (Studio MDHR) - 12 hrs
- Helldivers II (Arrowhead) - 0 hrs
```

Delete a game

Enter 4 to delete a game. Here is an example input for deleting a game:

```
Choice: 4
Title of game to delete: Helldivers II
Game deleted.
```

This should be reflected in your library:

```
Choice: 1

--- Your Library ---
- Elden Ring (FromSoftware) - 444 hrs
- Stardew Valley (ConcernedApe) - 2 hrs
- Cuphead (Studio MDHR) - 12 hrs
```

Search by title

Enter 5 to search for games by title:

```
Choice: 5
Title (partial OK): Cup
- Cuphead (2017-09-29) - Studio MDHR - PC - rating 8
```

Search by genre

Enter 6 to search for games by genre:

```
Choice: 6
Genre name: FPS
- Doom Eternal (id Software)
```

Search by tag

Enter 7 to search games by tag:

```
Choice: 7
Tag name: fun
- Elden Ring (FromSoftware)
- Hollow Knight (Team Cherry)
- Dark Souls III (FromSoftware)
- Cuphead (Studio MDHR)
```

View wishlist

Enter 8 to view your wishlist:

```
Choice: 8

--- Your Wishlist ---
- Celeste (2018-01-25)
- Doom Eternal (2020-03-20)
- Cuphead (2017-09-29)
```

Add to wishlist

Enter 9 to add to your wishlist:

```
Choice: 9
Title of game to add to wishlist: Stardew Valley
Added to wishlist.
```

This will be reflected in your wishlist

```
Choice: 8

--- Your Wishlist ---
- Celeste (2018-01-25)
- Stardew Valley (2016-02-26)
- Doom Eternal (2020-03-20)
- Cuphead (2017-09-29)
```

Remove from wishlist

Enter 10 to remove a game from your wishlist:

```
Choice: 10
Title of game to remove from wishlist: Stardew Valley
Removed from wishlist (if it was there).
```

This will be reflected in your wishlist

```
Choice: 8

--- Your Wishlist ---
- Celeste (2018-01-25)
- Doom Eternal (2020-03-20)
- Cuphead (2017-09-29)
```

View achievements

Enter 11 to view your achievements:

You must enter a game to view your achievements in,

```
Choice: 11
Title of game to view achievements for: Elden Ring

--- Achievements ---
- First Boss Defeated
```

Add achievement

Enter 12 to add an achievement:

You must select the game and achievement you want to 'earn',

```
Choice: 12
Title of game: Elden Ring
Achievement name: Elden Lord
Achievement 'Elden Lord' earned!
```

Assign tag to game

Enter 13 to assign a tag to a game:

You must select the game and tag name you want to assign,

```
Choice: 13
Title of game: Elden Ring
Tag name: casual
Tag 'casual' assigned.
```

Assign genre to game

Enter 14 to assign a genre to a game:

You must select the game and genre name you want to assign,

```
Choice: 14
Title of game: Elden Ring
Genre name: platformer
Genre 'platformer' assigned.
```

Set game as digital

Enter 15 to set a game as digital:

You must enter the name, file size, and store link of the game you want to add as digital,

```
Choice: 15
Title of game: Elden Ring
File size (in GB or whatever unit you use): 14
Store link: https://Elden\_Ring.com
Game set as digital.
```

Set game as physical

Enter 16 to set a game as physical:

You must enter the game name, physical condition, and shelf location to add as physical,

```
Choice: 16
Title of game: Elden Ring
Condition (e.g. great, good, bad): great
Shelf location: B1
Game set as physical.
```

View total playtime

Enter 17 to view total playtime across all games:

```
Choice: 17
Zach has played 912 hours total.
```

Logout

Enter 18 to logout, this will bring you back to the sign in screen with your original options available,

```
Choice: 18
Logging out...

--- Not Steam ---
1. Login
2. Register
3. Quit
Choice: █
```

Graphical User Interface:

To launch the GUI, navigate to the folder where you cloned the project repository. For example, my cloned folder is in ~/repos

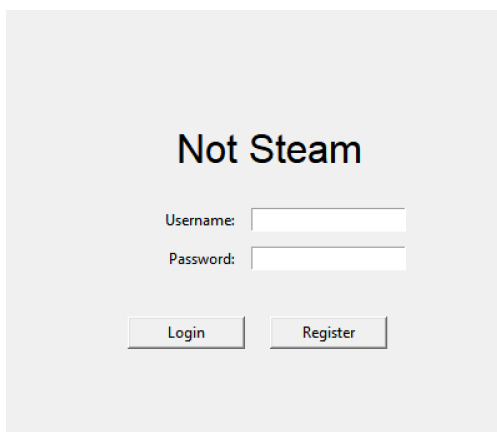
```
maddoxreed@MaddoxPC:~/repos/2300-Project$ ls
__pycache__  database.py  gui.py  main.py  notsteam.db  setup.sql
```

For Windows, use Powershell to navigate to the correct folder. For Ubuntu and Mac, use the Terminal.

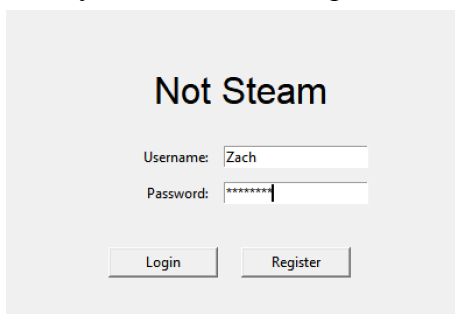
Launch gui.py with the following command:

```
python3 gui.py
```

The GUI should appear on screen as shown:

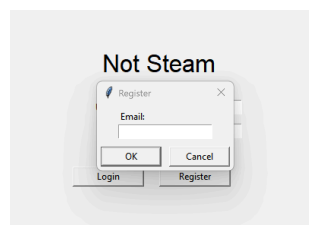


Enter your username and password in their respective boxes,



Press "Login"

- Note: If this is your first time using the application, you will need to enter your information like you were going to sign in, but click "Register" instead of "Login"

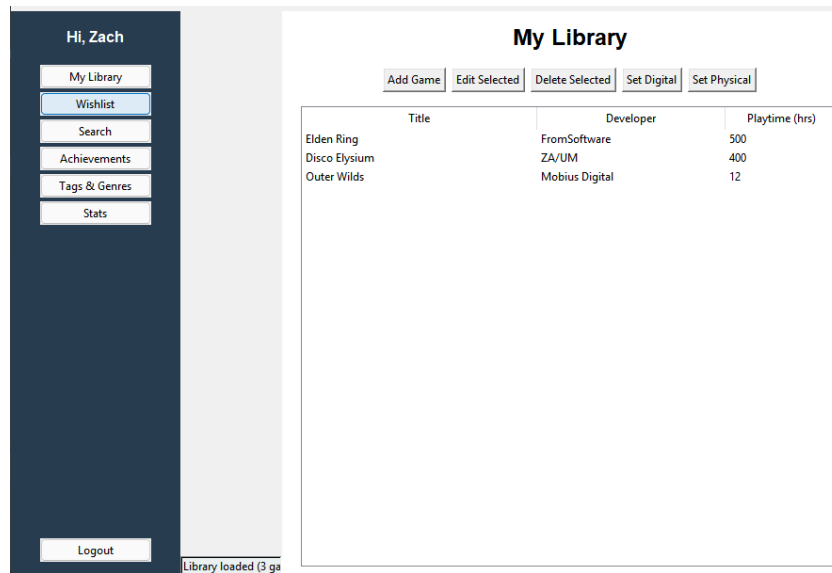


- This will open the email window above, prompting you to enter your email address.
- After you have entered your email address, press “OK”, then a message should appear to prompt you to press “Login”

This will now bring you to the My Library page.

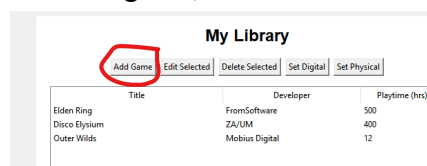
View my library

If you just registered, your library will be empty, but for the purposes of this guide we will use my pre-filled library.



Add a game

To add a game, click “Add Game” in the My Library button panel

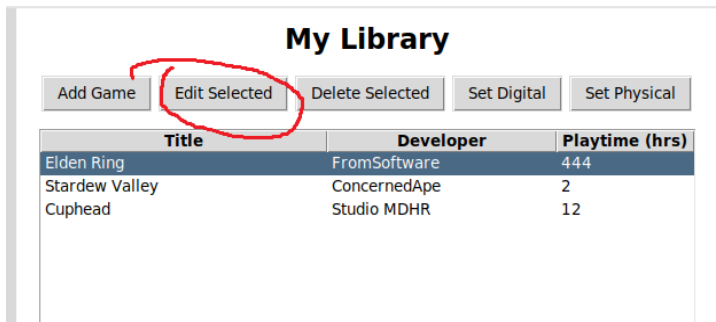


You will then be prompted to enter the game’s details as so:

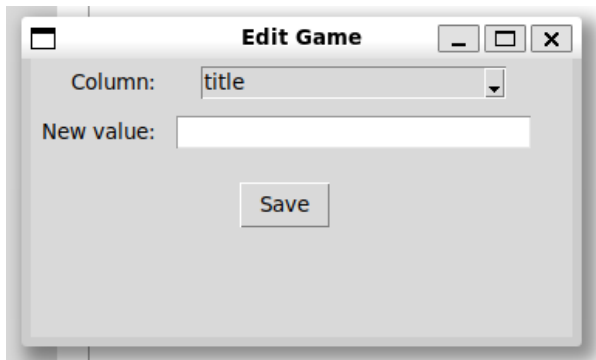
Click Add and your game will be in your library.

Edit a game

To edit a game, select a game in your library and then click “Edit Selected” in the My Library button panel

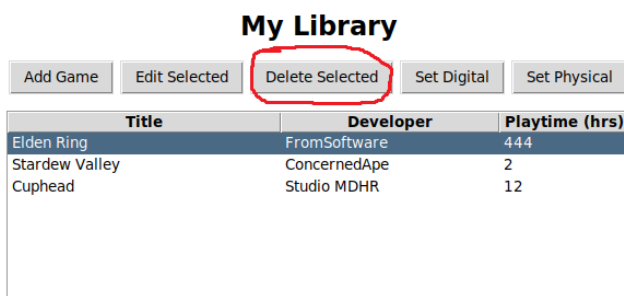


In the popup window, you can select the attribute you would like to edit, and you can assign the new value of the attribute. When you're done, click Save

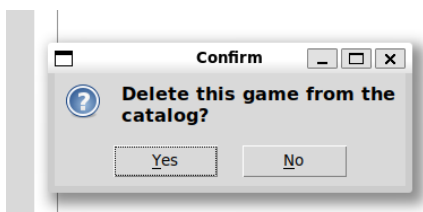


Delete a game

To delete a game, select a game in your library and then click “Delete Selected” in the My Library button panel

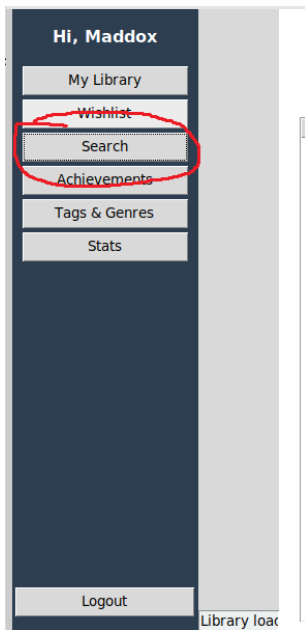


A window will appear asking you to confirm the deletion. If you would like to delete the game, select Yes



Search by title

To search for games by title, navigate to the Search tab



Leave the “Search by” option as the default, then type in the name of a game and click Search

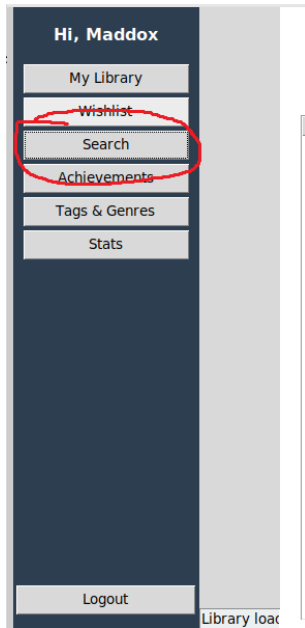
Search Games

Search by:

Title	Developer
Cuphead	Studio MDHR

Search by genre

To search for games by genre, navigate to the Search tab



Open the “Search by” dropdown and select genre, then type a game genre and click Search

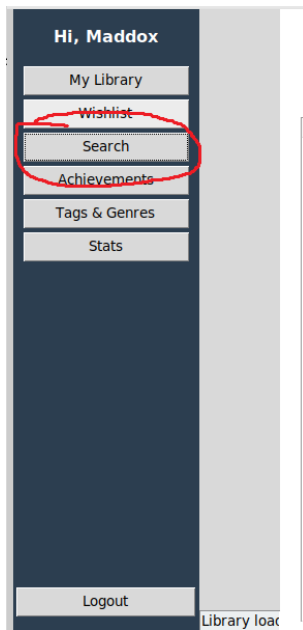
Search Games

Search by: genre

Title	Developer
Elden Ring	FromSoftware

Search by tag

To search for games by tag, navigate to the Search tab



Open the “Search by” dropdown and select tag, then type a game tag and click Search

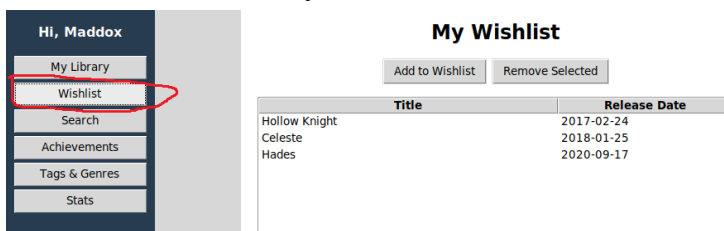
Search Games

Search by: tag

Title	Developer
Stardew Valley	ConcernedApe

View wishlist

Select Wishlist to view your wishlist



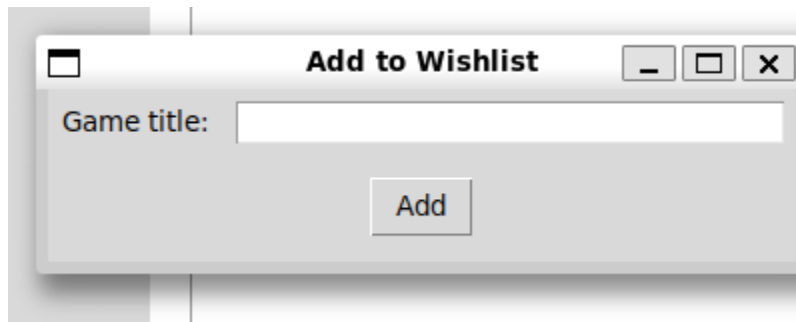
Add to wishlist

Select Wishlist to view your wishlist



On the “My Wishlist” screen, select “Add to Wishlist”

A popup window will appear. In this window, enter the name of the game you want to add and then click Add



This game will now appear in your wishlist

Remove from wishlist

Select Wishlist to view your wishlist

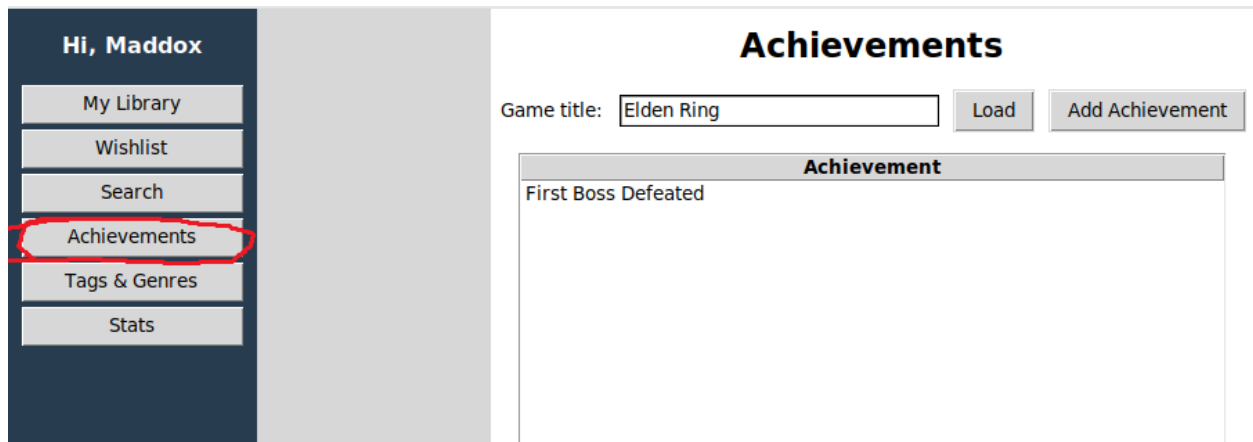


On the “My Wishlist” screen, select a game and then click “Remove Selected”

The game will no longer be displayed in your wishlist

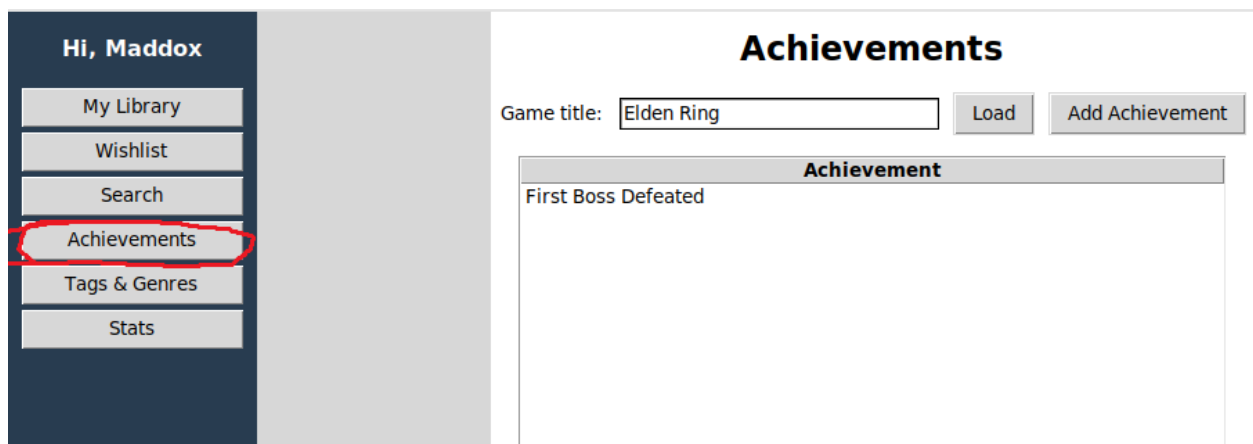
View achievements

Click Achievements on the left bar, then type the game's title you want. The achievements will then appear

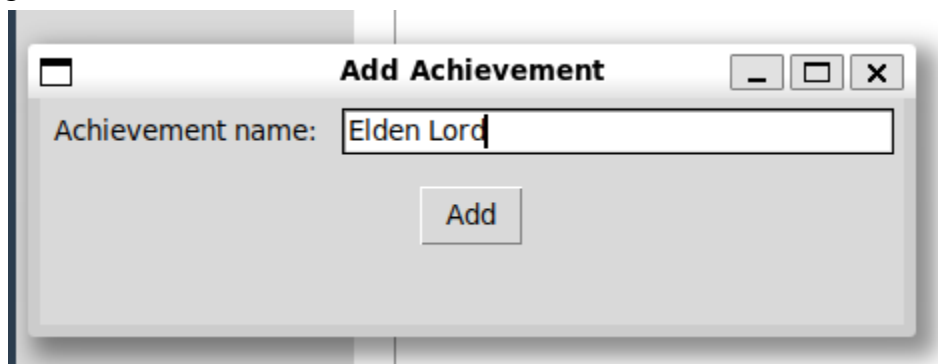


Add achievement

Click Achievements on the left bar, then type the game's title you want. The achievements will then appear



Then select “Add Achievement”. A new window will appear. Type in the name of the achievement you want to add. Click Add, and the achievement will be added to the selected game



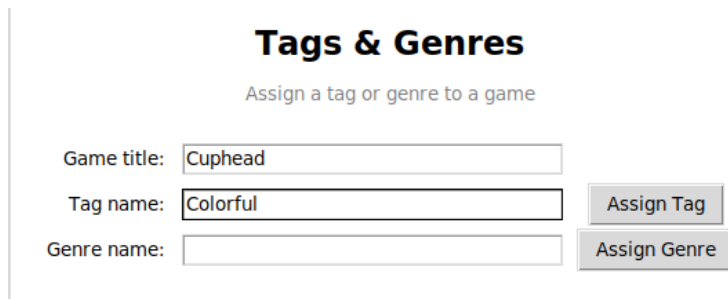
Assign tag to game

Click Tags & Genres on the left bar



The screenshot shows a user interface with a dark blue sidebar on the left and a light gray main area on the right. The sidebar contains a header 'Hi, Maddox' and a list of menu items: 'My Library', 'Wishlist', 'Search', 'Achievements', 'Tags & Genres', and 'Stats'. The 'Tags & Genres' item is highlighted with a red circle. The main area has a title 'Tags & Genres' and a subtitle 'Assign a tag or genre to a game'. It contains three input fields: 'Game title:', 'Tag name:', and 'Genre name:'. To the right of the 'Tag name:' field is a button labeled 'Assign Tag', and to the right of the 'Genre name:' field is a button labeled 'Assign Genre'.

In the Tags & Genres window, type in the title of the game you want to assign a tag to. Then, in the tag field, type the tag you want to add. Then click “Assign Tag”



The screenshot shows the 'Tags & Genres' window with the subtitle 'Assign a tag or genre to a game'. The 'Game title:' field is filled with 'Cuphead'. The 'Tag name:' field is filled with 'Colorful'. The 'Genre name:' field is empty. The 'Assign Tag' button is highlighted.

Assign genre to game

Click Tags & Genres on the left bar

Hi, Maddox

My Library

Wishlist

Search

Achievements

Tags & Genres

Stats

Tags & Genres

Assign a tag or genre to a game

Game title:

Tag name:

Genre name:

In the Tags & Genres window, type in the title of the game you want to assign a genre to. Then, in the genre field, type the genre you want to add. Then click “Assign Genre”

Tags & Genres

Assign a tag or genre to a game

Game title:

Tag name:

Genre name:

Set game as digital

Select a game in your library view, then click the Set Digital button

My Library

Title	Developer	Playtime (hrs)
Elden Ring	FromSoftware	444
Stardew Valley	ConcernedApe	2
Cuphead	Studio MDHR	12

In the following window, specify the file size and link to the store where the game is sold, then click Save

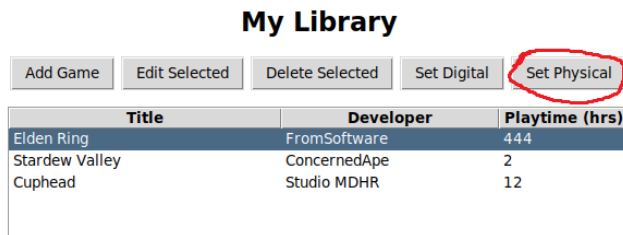
Set Digital

File size:

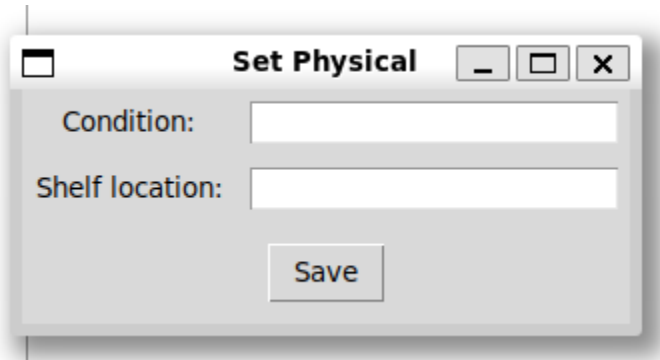
Store link:

Set game as physical

Select a game in your library view, then click the Set Physical button

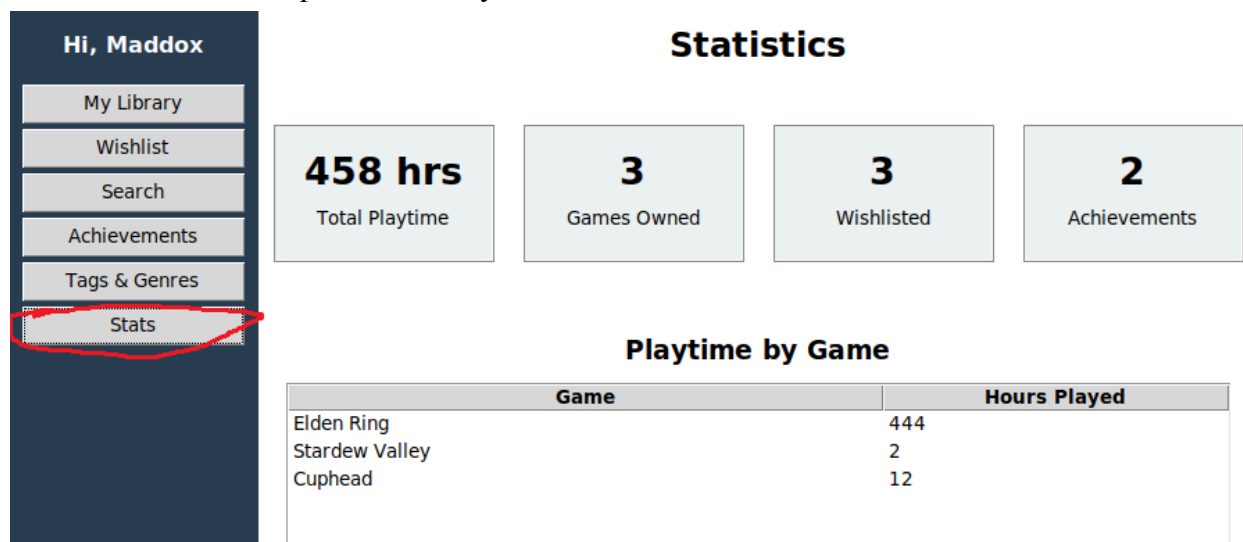


In the following window, specify the condition of the game and where the game is stored, then click Save



View statistics

Click Stats on the side panel to view your statistics



Logout

Click Logout on the bottom left of the window. This action will take you back to the login screen

Hi, Maddox

- My Library
- Wishlist
- Search
- Achievements
- Tags & Genres
- Stats
- Logout**

Library loaded (

My Library

Add Game Edit Selected Delete Selected Set Digital Set Physical

Title	Developer	Playtime (hrs)
Elden Ring	FromSoftware	444
Stardew Valley	ConcernedApe	2
Cuphead	Studio MDHR	12

To leave the GUI, simply click the X button at the top right of the window